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Storage system and computing system including the same

Abstract

A storage system includes a first storage device including a first write buffer and a first nonvolatile memory device, and a second storage device including a second write buffer and a second nonvolatile memory device. The first storage device is configured to, in response to a determination that a use buffer size of the first write buffer is greater than a first reference buffer size when the first storage device receives write data from a host, transfer the write data to the second storage device, and the second storage device is configured to store the write data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 USC § 119 to Korean Patent Application No. 10-2022-0002654, filed on Jan. 7, 2022 in the Korean Intellectual Property Office (KIPO), the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

1. Technical Field

(2) Example embodiments relate generally to semiconductor integrated circuits, and more particularly to a storage system including a nonvolatile memory device, and a computing system including the storage system.

2. Description of the Related Art

(3) A storage device including a nonvolatile memory device may perform a lazy write operation that first writes write data received from a host to a write buffer and then writes the write data stored in the write buffer to the nonvolatile memory device. Since a speed of a data write operation for the write buffer is faster than a speed of a data write operation for the nonvolatile memory device, the storage device may rapidly respond to a write command from the host by the lazy write operation.

SUMMARY

(4) Some example embodiments of the present inventive concepts provides a storage system having improved write performance. Such a storage system may have improved performance based on being configured to provide a solution to an issue where all storage space of a write buffer of a storage device stores write data and/or where the write buffer has no free buffer wherein a data write operation that writes new write data to the write buffer cannot be performed until the write data stored in the write buffer are written to a nonvolatile memory device.

(5) Some example embodiments of the present inventive concepts provides a computing system including a storage system having improved write performance.

(6) According to some example embodiments, a storage system may include a first storage device including a first write buffer and a first nonvolatile memory device, and a second storage device including a second write buffer and a second nonvolatile memory device. The first storage device may be configured to, in response to a determination that a use buffer size of the first write buffer is greater than a first reference buffer size when the first storage device receives write data from a host, transfer the write data to the second storage device, and the second storage device may be configured to store the write data.

(7) According to some example embodiments, a storage system includes a first storage device including a first write buffer, a mapping table and a first nonvolatile memory device, and a second storage device including a second write buffer and a second nonvolatile memory device. The first storage device may be configured to compare a use buffer size of the first write buffer with a first reference buffer size in response to a determination that the first storage device receives write data from a host. The first storage device may be configured to, in in response to a determination that the use buffer size of the first write buffer is greater than the first reference buffer size, transfer a buffer status request signal to the second storage device, and the second storage device is configured to transfer a buffer status response including a remaining buffer size of the second write buffer to the first storage device in response to the buffer status request signal. The first storage device may be configured to directly transfer the write data to the second storage device through P2P communication, the second storage device is configured to store the write data in the second write buffer, and the first storage device is configured to change a physical address for the write data in the mapping table to an address of the second storage device. The first storage device may be configured to, in response to a determination that the use buffer size of the first write buffer is decreased to less than a second reference buffer size, request the write data to the second storage device with the address of the second storage device, the second storage device is configured to

directly transfer the write data to the first storage device through the P2P communication, the first storage device is configured to store the write data in the first write buffer, and the first storage device is configured to write the write data stored in the first write buffer to the first nonvolatile memory device.

(8) According to some example embodiments, a computing system includes a storage system, and a host configured to store data in the storage system. The storage system includes a first storage device including a first write buffer and a first nonvolatile memory device, and a second storage device including a second write buffer and a second nonvolatile memory device. The first storage device may be configured to, in in response to a determination that a use buffer size of the first write buffer is greater than a first reference buffer size when the first storage device receives write data from the host, transfer the write data to the second storage device, and the second storage device is configured to store the write data.

(9) In a storage system and a computing system according to some example embodiments, in example embodiments where a use buffer size of a first write buffer of a first storage device is greater than a reference buffer size when the first storage device receives write data from a host, the first storage device may transfer the write data to a second storage device, and the second storage device may store the write data. Accordingly, the first storage device may not wait until the first write buffer becomes empty, and may rapidly respond to a write command from the host, thereby improving a write speed and write performance of the storage system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Illustrative, non-limiting example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings.

(2) FIG. **1** is a block diagram illustrating a computing system including a storage system according to some example embodiments.

(3) FIG. **2** is a flowchart for describing an example of an operation of a storage system according to some example embodiments.

(4) FIG. **3** is a diagram for describing an example where a use buffer size of a first write buffer is greater than a first reference buffer size according to some example embodiments.

(5) FIGS. **4**A, **4**B, and **4**C are diagrams for describing an example where remaining buffer sizes of write buffers of other storage devices are checked according to some example embodiments.

(6) FIG. **5** is a diagram for describing an example where a first storage device transfers write data to a second storage device according to some example embodiments.

(7) FIG. **6** is a diagram for describing an example where a first storage device receives write data that are temporarily stored in a second storage device according to some example embodiments.

(8) FIGS. **7**A and **7**B are separate portions of a flowchart for describing an example of an operation of a storage system according to some example embodiments.

(9) FIG. **8** is a diagram for describing an example where a second storage device temporarily stores write data in a second nonvolatile memory device according to some example embodiments.

(10) FIG. **9** is a diagram illustrating an example of a first mapping table of a first storage device according to some example embodiments.

(11) FIG. **10** is a diagram for describing an example where a first storage device receives write data that are temporarily stored in a second nonvolatile memory device of a second storage device according to some example embodiments.

(12) FIGS. **11**A and **11**B are separate portions of a flowchart for describing an example of an operation of a storage system according to some example embodiments.

(13) FIG. **12** is a diagram for describing an example where a second storage device temporarily

stores write data in an over-provisioning region of a second nonvolatile memory device according to some example embodiments.
(14) FIG. **13** is a block diagram illustrating a computing system according to some example embodiments.
(15) FIG. **14** is a block diagram illustrating an example of a nonvolatile memory device included in a storage device according to some example embodiments.
(16) FIG. **15** is a perspective view illustrating an example of a memory block included in a memory cell array of a nonvolatile memory device of FIG. **14** according to some example embodiments.
(17) FIG. **16** is a circuit diagram illustrating an equivalent circuit of a memory block described with reference to FIG. **15** according to some example embodiments.
(18) FIG. **17** is a cross-sectional view of a nonvolatile memory device included in a storage device according to some example embodiments.

DETAILED DESCRIPTION

(19) Various example embodiments will be described more fully with reference to the accompanying drawings, in which example embodiments are shown. The present inventive concepts may, however, be embodied in many different forms and should not be construed as limited to the example embodiments set forth herein. Like reference numerals refer to like elements throughout this application.
(20) It will be understood that when an element such as a layer, film, region, or substrate is referred to as being “on” another element, it may be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present. It will further be understood that when an element is referred to as being “on” another element, it may be above or beneath or adjacent (e.g., horizontally adjacent) to the other element.
(21) It will be understood that elements and/or properties thereof (e.g., structures, surfaces, directions, or the like), which may be referred to as being “perpendicular,” “parallel,” “coplanar,” or the like with regard to other elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) may be “perpendicular,” “parallel,” “coplanar,” or the like or may be “substantially perpendicular,” “substantially parallel,” “substantially coplanar,” respectively, with regard to the other elements and/or properties thereof.
(22) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially perpendicular” with regard to other elements and/or properties thereof will be understood to be “perpendicular” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “perpendicular,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a. tolerance of ±10%).
(23) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially parallel” with regard to other elements and/or properties thereof will be understood to be “parallel” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “parallel,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a. tolerance of ±10%).
(24) Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially coplanar” with regard to other elements and/or properties thereof will be understood to be “coplanar” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “coplanar,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a. tolerance of ±10%)).
(25) It will be understood that elements and/or properties thereof may be recited herein as being “the same” or “equal” as other elements, and it will be further understood that elements and/or

properties thereof recited herein as being "identical" to, "the same" as, or "equal" to other elements may be "identical" to, "the same" as, or "equal" to or "substantially identical" to, "substantially the same" as or "substantially equal" to the other elements and/or properties thereof. Elements and/or properties thereof that are "substantially identical" to, "substantially the same" as or "substantially equal" to other elements and/or properties thereof will be understood to include elements and/or properties thereof that are identical to, the same as, or equal to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances. Elements and/or properties thereof that are identical or substantially identical to and/or the same or substantially the same as other elements and/or properties thereof may be structurally the same or substantially the same, functionally the same or substantially the same, and/or compositionally the same or substantially the same.

(26) It will be understood that elements and/or properties thereof described herein as being "substantially" the same and/or identical encompasses elements and/or properties thereof that have a relative difference in magnitude that is equal to or less than 10%. Further, regardless of whether elements and/or properties thereof are modified as "substantially," it will be understood that these elements and/or properties thereof should be construed as including a manufacturing or operational tolerance (e.g., ±10%) around the stated elements and/or properties thereof.

(27) When the terms "about" or "substantially" are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of ±10% around the stated numerical value. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

(28) While the term "same," "equal" or "identical" may be used in description of some example embodiments, it should be understood that some imprecisions may exist. Thus, when one element is referred to as being the same as another element, it should be understood that an element or a value is the same as another element within a desired manufacturing or operational tolerance range (e.g., ±10%).

(29) When the terms "about" or "substantially" are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical value. Moreover, when the words "about" and "substantially" are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as "about" or "substantially," it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical values or shapes. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

(30) As described herein, when an operation is described to be performed "by" performing additional operations, it will be understood that the operation may be performed "based on" the additional operations, which may include performing said additional operations alone or in combination with other further additional operations.

(31) As described herein, an element that is described to be "spaced apart" from another element, in general and/or in a particular direction (e.g., vertically spaced apart, laterally spaced apart, etc.) and/or described to be "separated from" the other element, may be understood to be isolated from direct contact with the other element, in general and/or in the particular direction (e.g., isolated from direct contact with the other element in a vertical direction, isolated from direct contact with the other element in a lateral or horizontal direction, etc.). Similarly, elements that are described to be "spaced apart" from each other, in general and/or in a particular direction (e.g., vertically spaced apart, laterally spaced apart, etc.) and/or are described to be "separated" from each other, may be understood to be isolated from direct contact with each other, in general and/or in the particular direction (e.g., isolated from direct contact with each other in a vertical direction, isolated from

direct contact with each other in a lateral or horizontal direction, etc.).
(32) FIG. **1** is a block diagram illustrating a computing system including a storage system according to some example embodiments.
(33) Referring to FIG. **1**, a computing system **100** may include a storage system **200** including a plurality of storage devices **220**, **240** and **260**, and a host **120** storing data in the storage system **200**.
(34) In some example embodiments, the computing system **100** may be any computing system, such as a personal computer (PC), a server computer, a data center, a workstation, a digital television (TV), a set-top box, etc. In some example embodiments, the computing system **100** may be any mobile system, such as a mobile phone, a smart phone, a tablet computer, a laptop computer, a personal digital assistant (PDA), a portable multimedia player (PMP), a digital camera, a camcorder, a portable game console, a music player, a video player, a navigation device, a wearable device, an internet of things (IoT) device, an internet of everything (IoE) device, an e-book reader, a virtual reality (VR) device, an augmented reality (AR) device, a robotic device, a drone, etc.
(35) The host **120** may include a host processor **140** and a host memory **160**. The host processor **140** may transfer data stored in the host memory **160** as write data to the storage system **200**, or may store read data read from the storage system **200** to the host memory **160**.
(36) In some example embodiments, the host processor **140** and the host memory **160** may be implemented as separate semiconductor chips. In some example embodiments, the host processor **140** and the host memory **160** may be integrated in the same semiconductor chip. As an example, the host processor **140** may be any one of a plurality of modules included in an application processor (AP). The AP may be implemented as a System on Chip (SoC). Further, the host memory **160** may be an embedded memory included in the AP, or a nonvolatile memory device (NVM) or a memory module located outside the AP.
(37) The storage system **200** may be connected to the host **120** through an interface bus **180**. In some example embodiments, the interface bus **180** may be, but not be limited to, a peripheral component interconnection express (PCIe) bus or a nonvolatile memory express (NVMe) bus. In some example embodiments, the interface bus **180** may be, but not be limited to, an advanced technology attachment (ATA) bus, a serial ATA (SATA) bus, an external SATA (e-SATA) bus, a small computer small interface (SCSI) bus, a serial attached SCSI (SAS) bus, a peripheral component interconnection (PCI) bus, an IEEE 1394 bus, a universal serial bus (USB), a secure digital (SD) card bus, a multi-media card (MMC) bus, a universal flash storage (UFS) bus, an embedded MMC (eMMC) bus, an embedded UFS (eUFS) bus, a compact flash (CF) card bus, or the like.
(38) The storage system **200** may include N or more storage devices **220**, **240** and **260**, where N is an integer greater than 1. In some example embodiments, each storage device **220**, **240** and **260** may be a solid state drive (SSD) device. For example, each storage device **220**, **240** and **260** may be an SSD device that conforms to an NVMe standard. In some example embodiments, each storage device **220**, **240** and **260** may be a UFS device, a MMC device or an eMMC device. In some example embodiments, each storage device **220**, **240** and **260** may be an SD card, a micro SD card, a memory stick, a chip card, a USB card, a smart card, a CF card, or the like.
(39) Each storage device **220**, **240** and **260** may include a write buffer **222** and **242** and a nonvolatile memory device **224** and **244**. For example, a first storage device **220** may include a first write buffer **222** and a first nonvolatile memory device **224**, and a second storage device **240** may include a second write buffer **242** and a second nonvolatile memory device **244**.
(40) Each write buffer **222** and **242** may temporarily store data to be written to a corresponding nonvolatile memory device **224** and **244**. For example, if a write command that requests writing of write data is received from the host **120** (e.g., in response to a determination that a write command that requests writing of write data is received from the host **120**), the first storage device **220** may

temporarily store the write data received from the host **120** in the first write buffer **222**, and may write the write data stored in the first write buffer **222** to the first nonvolatile memory device **224**. A speed of a data write operation for each write buffer **222** and **242** may be faster than a speed of a data write operation for each nonvolatile memory device **224** and **244**, and thus each storage device **220**, **240** and **260** may rapidly respond to the write command by using the write buffer **222** and **242**.

(41) In some example embodiments, as described below with reference to FIG. **13**, each write buffer **222** and **242** may be included in a buffer memory. The buffer memory may include the write buffer **222** and **242** for temporarily storing the write data, a read buffer for temporarily storing read data, and an internal buffer for an internal operation within the storage device **220**, **240** and **260**. For example, a first buffer memory of the first storage device **220** may include the first write buffer **222**, a first read buffer and a first internal buffer, and a second buffer memory of the second storage device **240** may include the second write buffer **242**, a second read buffer and a second internal buffer. In some example embodiments, the buffer memory may be implemented with a volatile memory device, such as a dynamic random access memory (DRAM) or a static random access memory (SRAM). In some example embodiments, the buffer memory may be implemented with a nonvolatile memory device, such as a flash memory. For example, each nonvolatile memory device **224** and **244** may be implemented as a multi-level cell (MLC) memory, and the buffer memory may be implemented as a single level cell (SLC) memory faster than the MLC memory. Further, in some example embodiments, as illustrated in FIG. **13**, the buffer memory may be located inside a storage controller of each storage device **220**, **240** and **260**. In some example embodiments, the buffer memory may be implemented with a separate semiconductor chip located outside the storage controller. In still some example embodiments, a partial region (e.g., an SLC region) of each nonvolatile memory device **224** and **244** may be used as the buffer memory.

(42) Each nonvolatile memory device **224** and **244** may store the write data received through a corresponding write buffer **222** and **242** from the host **120**. Further, data stored in each nonvolatile memory device **224** and **244** may be provided as read data through the read buffer to the host **120**. In some example embodiments, each nonvolatile memory device **224** and **244** may be implemented with, but not limited to, a NAND flash memory. In some example embodiments, each nonvolatile memory device **224** and **244** may be implemented with an electrically erasable programmable read-only memory (EEPROM), a phase change random access memory (PRAM), a resistive random access memory (RRAM), a nano floating gate memory (NFGM), a polymer random access memory (PoRAM), a magnetic random access memory (MRAM), a ferroelectric random access memory (FRAM), or the like.

(43) In the storage system **200** according to some example embodiments, each storage device (e.g., the first storage device **220**) may use write buffers (e.g., the second write buffer **242**) of other storage devices (e.g., the second storage device **240**) when its write buffer (e.g., the first write buffer **222**) is full or is used by more than a threshold. For example, if the first storage device **220** receives the write data from the host **120** (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220**, that the first storage device **220** receives the write data from the host **120**), the first storage device **220** may compare a use buffer size of the first write buffer **22** with a first reference buffer size. Here, each write buffer **222** and **242** may include a plurality of buffers, and the use buffer size may represent a size of buffers storing (effective) data among the plurality of buffers. The use buffer size may correspond to a value calculated by subtracting a size of free buffers from a total size of the plurality of buffers. Further, the first reference buffer size may be a maximum buffer size (or the total size of the plurality of buffers) of each write buffer **222** and **242**, or may have a value less than the maximum buffer size. In example embodiments where the use buffer size of the first write buffer **222** is greater than the first reference buffer size, the first storage device may check remaining buffer sizes or free buffer sizes of the write buffers **242** of other storage device **240** and **260**.

(44) In some example embodiments, before (e.g., prior to) transferring the write data to one of the other storage devices **240** and/or **260**, for example to the second storage device **240**, to check the remaining buffer size of the second write buffer **242** of the second storage device **240**, the first storage device **220** may transfer a buffer status request signal to the second storage device **240**, and the second storage device **240** may transfer a buffer status response including the remaining buffer size of the second write buffer **242** to the first storage device **220** in response to the buffer status request signal. In example embodiments where the second write buffer **242** of the second storage device **240** has an available space, or free buffers, the first storage device **220** may transfer the write data to the second storage device **240**.
(45) In some example embodiments, the first storage device **220** may directly transfer the write data to the second storage device **240** through peer-to-peer (P2P) communication. In general, in example embodiments where data of the first storage device **220** are transferred to the second storage device **240**, the data of the first storage device **220** may be stored through the host processor **140** in the host memory **160**, and the data stored in the host memory **160** may be transferred through the host processor **140** to the second storage device **240**. However, in the storage system **200** according to some example embodiments, the write data may be directly transferred from the first storage device **220** through the interface bus **180** to the second storage device **240** without participation of the host processor **140** or without being stored in the host memory **160**. In some example embodiments, each storage device **220**, **240** and **260** may be the SSD device that conforms to the NVMe standard, and the first storage device **220** may directly transfer the write data to the second storage device **240** through P2P communication using a controller memory buffer (CMB) and/or a persistent memory region (PMR). In some example embodiments, the first storage device **220** may directly transfer the write data to the second storage device **240** through P2P communication of a compute express link (CXL). In some example embodiments, the first storage device **220** may directly transfer the write data to the second storage device **240** through P2P communication of a cache coherent interconnect for accelerators (CCIX).
(46) The second storage device **240** may store the write data transferred through the P2P communication in the second write buffer **242**. In some example embodiments, the second storage device **240** may store the write data in the second write buffer **242** until the second storage device **240** transfers the write data to the first storage device **220**. In some example embodiments, the second storage device **240** may write the write data stored in the second write buffer **242** to (a normal region of) the second nonvolatile memory device **244**. In still some example embodiments, the second nonvolatile memory device **244** may include not only the normal region that is accessible by the host **120**, but also an over-provisioning region that is not accessible by the host **120**, and the second storage device **240** may write the write data stored in the second write buffer **242** to the over-provisioning region of the second nonvolatile memory device **244**.
(47) If (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220**, that) the first storage device **220** receives a read command for the write data from the host **120**, or if (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220**, that) the use buffer size of the first write buffer **222** is decreased to less than a second reference buffer size, the first storage device **220** may receive the write data from the second storage device **240**.
(48) For example, in example embodiments where the first storage device **220** receives the read command for the write data from the host **120** while the write data are stored (e.g., concurrently with the write data being stored) in the second storage device **240**, the first storage device **220** may request the write data to the second storage device **240** (e.g., transmit a request to the second storage device **240** to transmit the write data to the first storage device **220** such that the first storage device **220** receives the write data from the second storage device **240**) through the P2P communication (e.g., of CMR/PMR, CXL or CCIX), and the second storage device **240** may directly transfer the write data to the first storage device **220** through the P2P communication (e.g.,

of CMR/PMR, CXL or CCIX). The first storage device **220** may store the write data transferred through the P2P communication in the first read buffer, and may output the write data stored in the first read buffer to the host **120**.
(49) Further, for example, in example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220**, that) the use buffer size of the first write buffer **222** is decreased to less than the second reference buffer size, and/or the first storage device **220** is in an idle state (e.g., the first storage device **220** has little or no command to be processed), the first storage device **220** may request the write data to the second storage device **240** (e.g., transmit a request to the second storage device **240** to transmit the write data to the first storage device **220** such that the first storage device **220** receives the write data from the second storage device **240**) through the P2P communication. In some example embodiments, the second reference buffer size may be less than or equal to the first reference buffer size. The second storage device **240** may directly transfer the write data to the first storage device **220** through the P2P communication. The first storage device **220** may store the write data transferred through the P2P communication in the first write buffer **222**, and may write the write data stored in the first write buffer **222** to the first nonvolatile memory device **224**.
(50) In a conventional storage system, in example embodiments where a write buffer of each storage device is full or is used by more than a threshold, a data write operation that writes new write data to the write buffer cannot be performed until write data stored in the write buffer are written to a nonvolatile memory device. However, in the storage system **200** and the computing system **100** including the storage system **200** according to some example embodiments, even if the first write buffer **222** is full or is used by more than a threshold, the first storage device **220** may transfer the write data received from the host **120** to the second storage device **240**, and the second storage device **240** may store the write data. Accordingly, the first storage device **220** may not wait until the first write buffer **222** becomes empty, and may rapidly respond to the write command from the host **120**, thereby improving a write speed and write performance of the storage system **200** and thereby improving the functionality of the storage system **200** and any device, system, or the like that includes the storage system **200** (e.g., the computing system **100** shown in FIG. **1**).
(51) FIG. **2** is a flowchart for describing an example of an operation of a storage system according to some example embodiments, FIG. **3** is a diagram for describing an example where a use buffer size of a first write buffer is greater than a first reference buffer size according to some example embodiments, FIGS. **4**A, **4**B, and **4**C are diagrams for describing an example where remaining buffer sizes of write buffers of other storage devices are checked according to some example embodiments, FIG. **5** is a diagram for describing an example where a first storage device transfers write data to a second storage device according to some example embodiments, and FIG. **6** is a diagram for describing an example where a first storage device receives write data that are temporarily stored in a second storage device according to some example embodiments.
(52) Referring to FIGS. **2** and **3**, when (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220**, that) a first storage device **220***a* receives first write data WD**1** from a host **120***a* (step S**300**), the first storage device **220***a* may compare a use buffer size UBS of a first write buffer **222***a* with a first reference buffer size RBS**1** (step S**310**). For example, the host **120***a* may execute a first virtual machine VM**1** and a second virtual machine VM**2**, and may further execute a hypervisor **130***a* that operates or controls the first and second virtual machines VM**1** and VM**2**. In an example of FIG. **3**, the first storage device **220***a* may be allocated to the first virtual machine VM**1**, and a second storage device **240***a* may be allocated to the second virtual machine VM**2**. In example embodiments, the first virtual machine VM**1** may transfer or write the first write data WD**1** to the first storage device **220***a* directly or through the hypervisor **130***a*, and the second virtual machine VM**2** may transfer or write second write data WD**2** to the second storage device **240***a* directly or through the hypervisor **130***a*. In some example embodiments, the comparing at step S**310** is performed in response to a start of receiving

the first write data WD**1** from the host at step S**300**. In some example embodiments, the comparing at step S**310** is performed at least partially concurrently with the first write data WD**1** being received at step S**300**.

(53) In example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***a*, that) the use buffer size UBS of the first write buffer **222***a* is less than or equal to the first reference buffer size RBS**1** (step S**310**: NO), the first storage device **220***a* may store the first write data WD**1** in the first write buffer **222***a* (step S**380**), may check a first physical address corresponding to a first logical address for the first write data WD**1** in a first mapping table **226***a*, and may write the first write data WD**1** stored in the first write buffer **222***a* to a first memory block having the first physical address within a first nonvolatile memory device (hereinafter, "NVM1") **224***a* (step S**390**). As described below with reference to FIG. **13**, each storage device **220***a* and **240***a* may include a mapping table **226***a* and **246***a* that stores physical addresses of a corresponding nonvolatile memory device **224***a* and **244***a* corresponding to logical addresses received from the host **120***a*. Further, in example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the second storage device **240***a*, that) a use buffer size of a second write buffer **242***a* is less than or equal to the first reference buffer size RBS**1**, the second storage device **240***a* may store the second write data WD**2** in the second write buffer **242***a*, may check a second physical address corresponding to a second logical address for the second write data WD**2** in a second mapping table **246***a*, and may write the second write data WD**2** stored in the second write buffer **242***a* to a second memory block having the second physical address within a second nonvolatile memory device (hereinafter, "NVM2") **244***a*.

(54) In some example embodiments, in example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***a*, that) the use buffer size UBS of the first write buffer **222***a* is greater than the first reference buffer size RBS**1** (step S**310**: YES) (for example, a determination at step S**310** that the use buffer size UBS of the first write buffer **222***a* is greater than the first reference buffer size RBS**1**, where such determination at step S**310** may be performed in response to the first storage device **220***a* receiving the write data from WD**1** from the host **120***a* at step S**300**), the first write buffer **222***a* may check remaining buffer sizes of write buffers **242***a* of other storage devices **240***a* (step S**320** and step S**325**). In some example embodiments, as illustrated in FIG. **4**A, the first storage device **220***a* may broadcast a buffer status request signal BSRS to other storage devices **240***a* and **260***a* through an interface bus **180***a* (step S**320**). As illustrated in FIG. **4**B, the other storage devices **240***a* and **260***a* may transfer buffer status responses RES2 and RESN including the remaining buffer sizes of their write buffers **242***a* through the interface bus **180***a* in response to the buffer status request signal BSRS, respectively (step S**325**). For example, as illustrated in FIG. **4**C, the buffer status response RES2 of the second storage device **240***a* may include a total buffer size of the second write buffer **242***a* (e.g., may include information indicating a total buffer size of the second write buffer **242***a*), the remaining buffer size of the second write buffer **242***a*, and information about (e.g., information indicating) whether the second write buffer **242***a* is permitted to be used by another storage, or the first storage device **220***a*. FIG. **4**C illustrates an example where an N-th storage device **260***a* does not permit the first storage device **220***a* to use its write buffer, but the second storage device **240***a* permits the first storage device **220***a* to use the second write buffer **242***a*. The first storage device **220***a* may select a storage device to which the first write data WD**1** are to be transferred from the other storage devices **240***a* and **260***a* based on the buffer status responses RES2 and RESN. In some example embodiments, the first storage device **220***a* may select, but not limited to, a storage device having the largest remaining buffer size. Further, in some example embodiments, where the other storage devices **240***a* and **260***a* are incompatible with the first storage device **220***a*, the other storage devices **240***a* and **260***a* may not respond to the buffer status request signal BSRS.

(55) In example embodiments where the second storage device **240***a* is selected based on the buffer status responses RES2 and RESN, as illustrated in FIG. **5**, the first storage device **220***a* may directly transfer the first write data WD**1** to the second storage device **240***a* through P2P communication without participation of the host **120***a* (step S**330**). Accordingly, the first storage device **220***a* may not wait until the first write buffer **222***a* becomes empty, and may rapidly respond to the write command from the host **120**, thereby improving a write speed and write performance of the storage system **200** and thereby improving the functionality of the storage system **200** and any device, system, or the like that includes the storage system **200** (e.g., the computing system **100** shown in FIG. **1**). If the first write data WD**1** are received through the P2P communication, the second storage device **240***a* may store not only the second write data WD**2** in the second write buffer **242***a*, but also the first write data WD**1** received through the P2P communication in the second write buffer **242***a* (step S**410**). In some example embodiments, the second write data WD**2** stored in the second write buffer **242***a* may be written to the NVM2 **244***a*, but the second write buffer **242***a* may store the first write data WD**1** until the first write data WD**1** are transferred or returned to the first storage device **220***a*. If the first write data WD**1** are stored in the second storage device **240***a*, the first storage device **220***a* may change or update the first physical address corresponding to the first logical address for the first write data WD**1** in the first mapping table **226***a* from an original physical address (e.g., an address of a memory block within the NVM1 **224***a*) to an address of the second storage device **240***a* (step S**340**). For example, if the second storage device **240***a* stores the first write data WD**1** in the second write buffer **242***a*, the second storage device **240***a* may transfer a success response including a logical address within the second storage device **240***a* to the first storage device **220***a*, and the first storage device **220***a* may change or update the first physical address for the first write data WD**1** in the first mapping table **226***a* to the logical address received from the second storage device **240***a*.
(56) In example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***a*, that) the use buffer size UBS of the first write buffer **222***a* is greater than or equal to a second reference buffer size (step S**350**: NO), the first write data WD**1** may be retained in the second storage device **240***a*. In example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***a*, that) the first storage device **220***a* receives a read command for the first write data WD**1** from the host **120***a* while the first write data WD**1** are stored (e.g., concurrently with the first write data WD**1** being stored) in the second storage device **240***a*, the first storage device **220***a* may receive the first write data WD**1** from the second storage device **240***a*, may store the first write data WD**1** in a first read buffer of the first storage device **220***a*, may output the first write data WD**1** stored in the first read buffer to the host **120***a*.
(57) In some example embodiments, as illustrated in FIG. **6**, in example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***a*, that) the use buffer size UBS of the first write buffer **222***a* is decreased to less than the second reference buffer size RBS**2** (step S**350**: YES), and/or the first storage device **220***a* is in an idle state, the first storage device **220***a* may receive the first write data WD**1** from the second storage device **240***a* (step S**360** and step S**420**). In some example embodiments, the first storage device **220***a* may request the first write data WD**1** to the second storage device **240***a* based on the address of the second storage device **240***a* for the first write data WD**1** in the first mapping table **226***a* (step S**360**). The second storage device **240***a* may directly transfer the first write data WD**1** stored in the second write buffer **242***a* to the first storage device **220***a* through the P2P communication without participation of the host **120***a* (step S**420**).
(58) If (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***a*, that) the first write data WD**1** are received or returned through the P2P communication, the first storage device **220***a* may change or update the first physical

address corresponding to the first logical address for the first write data WD**1** in the first mapping table **226***a* from the address of the second storage device **240***a* to the original physical address, or the address of the memory block within the NVM1 **224***a* (step S**370**). Further, the first storage device **220***a* may store the first write data WD**1** received through the P2P communication in the first write buffer **222***a* (step S**380**), and may write the first write data WD**1** to the memory block within the NVM1 **224***a* based on the first physical address for the first write data WD**1** in the first mapping table **226***a* (step S**390**).

(59) FIGS. **7**A and **7**B are separate portions of a flowchart for describing an example of an operation of a storage system according to some example embodiments, FIG. **8** is a diagram for describing an example where a second storage device temporarily stores write data in a second nonvolatile memory device system according to some example embodiments, FIG. **9** is a diagram illustrating an example of a first mapping table of a first storage device system according to some example embodiments, and FIG. **10** is a diagram for describing an example where a first storage device receives write data that are temporarily stored in a second nonvolatile memory device of a second storage device system according to some example embodiments.

(60) An operation of a storage system illustrated in FIGS. **7**A and **7**B may be similar to an operation of a storage system illustrated in FIG. **2**, except that a second storage device **240***b* may store write data WD received through P2P communication from a first storage device **220***b* in an NVM2 **244***b*.

(61) Referring to FIGS. **7**A, **7**B and **8**, when (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***b*, that) the first storage device **220***b* receives the write data WD from a host (step S**500**), the first storage device **220***b* may compare a use buffer size of a first write buffer **222***b* with a first reference buffer size (step S**510**). In some example embodiments, the comparing at step S**510** is performed in response to a start of receiving the write data WD from the host at step S**500**. In some example embodiments, the comparing at step S**510** is performed at least partially concurrently with the write data WD being received at step S**500**. In example embodiments where (e.g., in response to a determination, for example by some portion of the storage system **200** such as the first storage device **220***b*, that) the use buffer size of the first write buffer **222***b* is less than or equal to the first reference buffer size (step S**510**: NO), where such determination at step S**510** may be made when (e.g., in response to) the first storage device **220***b* receives the write data WD from the host at step S**500**, the first storage device **220***b* may store the write data WD in the first write buffer **222***b* (step S**580**), and may write the write data WD**1** stored in the first write buffer **222***b* to an NVM1 **224***b* (step S**590**). In some example embodiments, in example embodiments where the use buffer size of the first write buffer **222***b* is greater than the first reference buffer size (step S**510**: YES), where such determination at step S**510** may be made when (e.g., in response to) the first storage device **220***b* receives the write data WD from the host at step S**500**, the first storage device **220***b* may broadcast a buffer status request signal to other storage devices **240***b* (step S**520**), and may receive buffer status responses including remaining buffer sizes of their write buffers **242***b* from the other storage devices **240***b* (step S**525**).

(62) In example embodiments where the second storage device **240***b* is selected based on the buffer status responses, as illustrated in FIG. **8**, the first storage device **220***b* may directly transfer the write data WD to the second storage device **240***b* through P2P communication without participation of the host (step S**530**). Accordingly, the first storage device **220***b* may not wait until the first write buffer **222***b* becomes empty, and may rapidly respond to the write command from the host, thereby improving a write speed and write performance of the storage system **200** and thereby improving the functionality of the storage system **200** and any device, system, or the like that includes the storage system **200** (e.g., the computing system **100** shown in FIG. **1**). If the write data WD are received through the P2P communication, the second storage device **240***b* may store the write data WD in a second write buffer **242***b* (step S**610**), and then may write the write data WD stored in the

second write buffer **242***b* to a memory block MB of the NVM2 **244***b* (step S**630**).
(63) Further, if the write data WD are stored in the second write buffer **242***b* (step S**610**), the second storage device **240***b* may check a logical address LA corresponding to an address (e.g., a physical address) of the memory block MB in a second mapping table **246***b*, and may transfer the logical address LA to the first storage device **220***b* (step S**620**). The first storage device **220***b* may change a physical address for the write data WD in a first mapping table **226***b* from an address of a memory block of the NVM1 **224***b* to the logical address LA received from the second storage device **240***b*.
(64) For example, as illustrated in FIG. **9**, the first mapping table **226***b* may store a plurality of logical addresses (e.g., logical block addresses) and a plurality of physical address (e.g., physical block addresses) corresponding to the logical addresses, and may further store location information representing (e.g., indicating) which one of a plurality of storage devices **220***b* and/or **240***b* store data corresponding to each pair of the logical and physical address. In an example of FIG. **9**, SSD**1** may represent the first storage device **220***b*, and SSD**2** may represent the second storage device **240***b*. Further, FIG. **9** illustrates an example where the write data WD having a logical address of '0x10' are stored in the second storage device **240***b*. If the write data WD are stored in the second storage device **240***b*, the first storage device **220***b* may change a physical address corresponding to the logical address of '0x10' to the logical address LA of '0x99' within the second storage device **240***b*.
(65) In example embodiments where the use buffer size of the first write buffer **222***b* is greater than or equal to a second reference buffer size (step S**550**: NO), the write data WD may be retained in the memory block MB of the NVM2 **244***b* of the second storage device **240***b*. In example embodiments where the use buffer size of the first write buffer **222***b* is decreased to less than the second reference buffer size (step S**550**: YES), as illustrated in FIG. **10**, the first storage device **220***b* may receive the write data WD from the second storage device **240***b* (step S**560**, step S**640** and step S**650**). In some example embodiments, the first storage device **220***b* may request the write data WD to the second storage device **240***b* with the logical address LA of the second storage device **240***b* stored in the first mapping table **226***b* (step S**560**). The second storage device **240***b* may store the write data WD in a second read buffer **248***b* by reading the write data WD from the memory block MB of the NVM2 **244***b* based on the address of the memory block MB of the NVM2 **244***b* corresponding to the logical address LA (step S**640**), and may transfer the write data WD stored in the second read buffer **248***b* to the first storage device **220***b* through the P2P communication (step S**650**).
(66) The first storage device **220***b* may change, update or recover the physical address for the write data WD in the first mapping table **226***b* to the address of the memory block of the NVM1 **224***b* (step S**570**), may store the write data WD received through the P2P communication in the first write buffer **222***b* (step S**580**), and may write the write data WD to the memory block of the NVM1 **224***b* (step S**590**). Thereafter, if the host transfers a read command that requests the write data WD, the first storage device **220***b* may store the write data WD in a first read buffer **228***b* by reading the write data WD from the memory block of the NVM1 **224***b*, and may provide the write data WD stored in the first read buffer **228***b* as read data to the host.
(67) FIGS. **11**A and **11**B are separate portions of a flowchart for describing an example of an operation of a storage system according to some example embodiments, and FIG. **12** is a diagram for describing an example where a second storage device temporarily stores write data in an over-provisioning region of a second nonvolatile memory device according to some example embodiments.
(68) An operation of a storage system illustrated in FIGS. **11**A and **11**B may be similar to an operation of a storage system illustrated in FIGS. **7**A and **7**B, except that a second storage device **240***c* may store write data WD received through P2P communication from a first storage device **220***c* in an over-provisioning region OPR of an NVM2 **244***c*.

(69) Referring to FIGS. **11**A, **11**B and **12**, each nonvolatile memory device **244***c* may include a normal region NR that is accessible by a host, and an over-provisioning region OPR that is not accessible by the host. For example, a mapping table **246***c* may store a logical address corresponding to a physical address of a memory block within the normal region NR, but not store a logical address corresponding to a physical address of a memory block within the over-provisioning region OPR. Thus, each nonvolatile memory device **244***c* may not provide the logical address for the over-provisioning region OPR to the host, and the host may not be able to access the over-provisioning region OPR. In some example embodiments, the over-provisioning region OPR may be a region for an internal operation (e.g., a wear-leveling operation, a garbage collection operation, or the like) of each storage device **220***c* and **240***c*.
(70) If the second storage device **240***c* receives the write data WD through the P2P communication from the first storage device **220***c* (step S**530**), the second storage device **240***c* may store the write data WD in a second write buffer **242***c* (step S**810**), and then may write the write data WD stored in the second write buffer **242***c* to a memory block within the over-provisioning region OPR of the NVM2 **244***c* (step S**840**). Further, the second storage device **240***c* may generate a logical address LA corresponding to an address of the memory block within the over-provisioning region OPR in a second mapping table **246***c* (step S**820**), and may transfer the logical address LA to the first storage device **220***c* (step S**830**). The first storage device **220***c* may change or update a physical address for the write data WD in a first mapping table to the logical address LA received from the second storage device **240***c* (step S**540**).
(71) Thereafter, the first storage device **220***c* may request the write data WD to the second storage device **240***c* with the logical address LA (step S**560**). The second storage device **240***c* may store the write data WD in a second read buffer **248***c* by reading the write data WD from the memory block within the over-provisioning region OPR based on the address of the memory block within the over-provisioning region OPR corresponding to the logical address LA in the second mapping table **246***c* (step S**850**), and may transfer the write data WD stored in the second read buffer **248***c* to the first storage device **220***c* through the P2P communication (step S**860**).
(72) FIG. **13** is a block diagram illustrating a computing system according to some example embodiments.
(73) Referring to FIG. **13**, a computing system may include a storage system **200** including a plurality of storage devices **220**, **240**, . . . , and a host **120** storing data in the storage system **200**. The host **120** may include a host processor **140** and a host memory **160**. Each storage device **220** may include a storage controller **223** and a nonvolatile memory device (hereinafter, "NVM") **224**. In some example embodiments, the NVM **224** may include a flash memory, and the flash memory may include a 2D NAND memory array or a 3D (or vertical) NAND (VNAND) memory array. In some example embodiments, the NVM **224** may include various other kinds of NVMs, such as, an MRAM, a spin-transfer torque MRAM, a conductive bridging RAM (CBRAM), an FRAM, PRAM, RRAM, and various other kinds of memories.
(74) The storage controller **223** may include a host interface **230**, a memory interface **232**, a central processing unit (CPU) **233** and a buffer memory **236**. The storage controller **223** may further include a flash translation layer (FTL) **234**, a packet manager **235**, an error correction code (ECC) engine **237** and an advanced encryption standard (AES) engine **238**. The storage controller **223** may further include a working memory (not shown) in which the FTL **234** is loaded, and the CPU **233** may execute the FTL **234** to control data write and read operations on the NVM **224**.
(75) The host interface **230** may transmit and receive packets to and from the host **120**. A packet transmitted from the host **120** to the host interface **230** may include a command or data to be written to the NVM **224**. A packet transmitted from the host interface **230** to the host **120** may include a response to the command or data read from the NVM **224**. The memory interface **232** may transmit data to be written to the NVM **224** to the NVM **224**, or may receive data read from the NVM **224**. The memory interface **232** may be configured to comply with a standard protocol,

such as Toggle or open NAND flash interface (ONFI).
(76) The FTL **234** may perform various functions, such as an address mapping operation, a wear-leveling operation, and a garbage collection operation. The address mapping operation may be an operation of converting a logical address received from the host **120** into a physical address used to actually store data in the NVM **224**. The wear-leveling operation may be a technique for reducing or preventing excessive deterioration of a specific block by allowing blocks of the NVM **224** to be uniformly used. As an example, the wear-leveling operation may be implemented using a firmware technique that balances erase counts of physical blocks. The garbage collection operation may be a technique for ensuring usable capacity in the NVM **224** by erasing an existing block after copying valid data of the existing block to a new block.
(77) The packet manager **235** may generate a packet according to a protocol of an interface, which consents to the host **120**, or parse various types of information from the packet received from the host **120**. In addition, the buffer memory **236** may include a write buffer WB that temporarily stores write data to be written to the NVM **224**, a read buffer RB that temporarily stores read data read from the NVM **224**, and an internal buffer for an internal operation. Although the buffer memory **236** may be a component included in the storage controller **223**, the buffer memory **236** may be outside the storage controller **223**.
(78) The ECC engine **237** may perform error detection and correction operations on read data read from the NVM **224**. For example, the ECC engine **237** may generate parity bits for write data to be written to the NVM **224**, and the generated parity bits may be stored in the NVM **224** together with write data. During the reading of data from the NVM **224**, the ECC engine **237** may correct an error in the read data by using the parity bits read from the NVM **224** along with the read data, and output error-corrected read data.
(79) The AES engine **238** may perform at least one of an encryption operation and/or a decryption operation on data input to the storage controller **223** by using a symmetric-key algorithm.
(80) FIG. **14** is a block diagram illustrating an example of a nonvolatile memory device included in a storage device according to some example embodiments.
(81) Referring to FIG. **14**, a nonvolatile memory device **300** may include a memory cell array **330**, and a control circuit that performs an operation for the memory cell array **330**. The control circuit may include a control logic circuitry **320**, a page buffer circuit **340**, a voltage generator **350** and a row decoder **360**. Although not shown in FIG. **14**, the nonvolatile memory device **300** may further include an interface circuitry **310**. In addition, the nonvolatile memory device **300** may further include column logic, a pre-decoder, a temperature sensor, a command decoder, and/or an address decoder.
(82) The control logic circuitry **320** may control all various operations of the nonvolatile memory device **300**. The control logic circuitry **320** may output various control signals in response to commands CMD and/or addresses ADDR from the interface circuitry **310**. For example, the control logic circuitry **320** may output a voltage control signal CTRL_vol, a row address X-ADDR, and a column address Y-ADDR.
(83) The memory cell array **330** may include a plurality of memory blocks BLK**1** to BLKz (here, z is a positive integer), each of which may include a plurality of memory cells. The memory cell array **330** may be connected to the page buffer circuit **340** through bitlines BL and be connected to the row decoder **360** through wordlines WL, string selection lines SSL, and ground selection lines GSL.
(84) In some example embodiments, the memory cell array **330** may include a 3D memory cell array, which includes a plurality of NAND strings. Each of the NAND strings may include memory cells respectively connected to wordlines vertically stacked on a substrate. The inventive concepts of U.S. Pat. Nos. 7,679,133; 8,553,466; 8,654,587; 8,559,235; and US Pat. Pub. No. 2011/0233648 are hereby incorporated by reference. In some example embodiments, the memory cell array **330** may include a 2D memory cell array, which includes a plurality of NAND strings arranged in a row

direction and a column direction.
(85) The page buffer circuit **340** may include a plurality of page buffers PB**1** to PBm (here, m is an integer greater than or equal to 3), which may be respectively connected to the memory cells through a plurality of bitlines BL. The page buffer circuit **340** may select at least one of the bitlines BL in response to the column address Y-ADDR. The page buffer circuit **340** may operate as a write driver or a sense amplifier according to an operation mode. For example, during a program operation, the page buffer circuit **340** may apply a bitline voltage corresponding to data to be programmed, to the selected bitline. During a read operation, the page buffer circuit **340** may sense current or a voltage of the selected bitline BL and sense data stored in the memory cell.
(86) The voltage generator **350** may generate various kinds of voltages for program, read, and erase operations based on the voltage control signal CTRL_vol. For example, the voltage generator **350** may generate a program voltage, a read voltage, a program verification voltage, and an erase voltage as a wordline voltage VWL.
(87) The row decoder **360** may select one of a plurality of wordlines WL and select one of a plurality of string selection lines SSL in response to the row address X-ADDR. For example, the row decoder **360** may apply the program voltage and the program verification voltage to the selected wordline WL during a program operation and apply the read voltage to the selected word line WL during a read operation.
(88) FIG. **15** is a perspective view illustrating an example of a memory block included in a memory cell array of a nonvolatile memory device of FIG. **14** according to some example embodiments.
(89) Referring to FIG. **15**, a memory block BLKi includes a plurality of cell strings (e.g., a plurality of vertical NAND strings) which are formed on a substrate in a three-dimensional structure (or a vertical structure). The memory block BLKi includes structures extending along first, second and third directions D**1**, D**2** and D**3**.
(90) A substrate **111** is provided. For example, the substrate **111** may have a well of a first type of charge carrier impurity (e.g., a first conductivity type) therein. For example, the substrate **111** may have a p-well formed by implanting a group **3** element such as boron (B). In particular, the substrate **111** may have a pocket p-well provided within an n-well. In some example embodiments, the substrate **111** has a p-type well (or a p-type pocket well). However, the conductivity type of the substrate **111** is not limited to p-type.
(91) A plurality of doping regions **311**, **312**, **313** and **314** arranged along the second direction D**2** are provided in/on the substrate **111**. This plurality of doping regions **311** to **314** may have a second type of charge carrier impurity (e.g., a second conductivity type) different from the first type of the substrate **111**. In some example embodiments of the inventive concepts, the first to fourth doping regions **311** to **314** may have n-type. However, the conductivity type of the first to fourth doping regions **311** to **314** is not limited to n-type.
(92) A plurality of insulation materials **112** extending along the first direction D**1** are sequentially provided along the third direction D**3** on a region of the substrate **111** between the first and second doping regions **311** and **312**. For example, the plurality of insulation materials **112** are provided along the third direction D**3**, being spaced by a specific distance. For example, the insulation materials **112** may include or may be formed of an insulation material such as an oxide layer.
(93) A plurality of pillars **113** penetrating the insulation materials along the third direction D**3** are sequentially disposed along the first direction D**1** on a region of the substrate **111** between the first and second doping regions **311** and **312**. For example, the plurality of pillars **113** penetrates the insulation materials **112** to contact the substrate **111**.
(94) In some example embodiments, each pillar **113** may include a plurality of materials. For example, a channel layer **114** of each pillar **113** may include or may be formed of a silicon material having a first conductivity type. For example, the channel layer **114** of each pillar **113** may include or may be formed of a silicon material having the same conductivity type as the substrate **111**. In some example embodiments of the inventive concepts, the channel layer **114** of each pillar **113**

includes or is formed of p-type silicon. However, the channel layer **114** of each pillar **113** is not limited to the p-type silicon.
(95) An internal material **115** of each pillar **113** includes an insulation material. For example, the internal material **115** of each pillar **113** may include or may be formed of an insulation material such as a silicon oxide. In some examples, the internal material **115** of each pillar **113** may include an air gap. The term “air” as discussed herein, may refer to atmospheric air, or other gases that may be present during the manufacturing process.
(96) An insulation layer **116** is provided along the exposed surfaces of the insulation materials **112**, the pillars **113**, and the substrate **111**, on a region between the first and second doping regions **311** and **312**. For example, the insulation layer **116** provided on surfaces of the insulation material **112** may be interposed between pillars **113** and a plurality of stacked first conductive materials **211**, **221**, **231**, **241**, **251**, **261**, **271**, **281** and **291**, as illustrated. In some examples, the insulation layer **116** need not be provided between the first conductive materials **211** to **291** corresponding to ground selection lines GSL (e.g., **211**) and string selection lines SSL (e.g., **291**). For example, the ground selection lines GSL are the lowermost ones of the stack of first conductive materials **211** to **291** and the string selection lines SSL are the uppermost ones of the stack of first conductive materials **211** to **291**.
(97) The plurality of first conductive materials **211** to **291** are provided on surfaces of the insulation layer **116**, in a region between the first and second doping regions **311** and **312**. For example, the first conductive material **211** extending along the first direction D**1** is provided between the insulation material **112** adjacent to the substrate **111** and the substrate **111**. In more detail, the first conductive material **211** extending along the first direction D**1** is provided between the insulation layer **116** at the bottom of the insulation material **112** adjacent to the substrate **111** and the substrate **111**.
(98) A first conductive material extending along the first direction D**1** is provided between the insulation layer **116** at the top of the specific insulation material among the insulation materials **112** and the insulation layer **116** at the bottom of a specific insulation material among the insulation materials **112**. For example, a plurality of first conductive materials **221** to **281** extending along the first direction D**1** are provided between the insulation materials **112** and it may be understood that the insulation layer **116** is provided between the insulation materials **112** and the first conductive materials **221** to **281**. The first conductive materials **211** to **291** may be formed of a conductive metal, but in some example embodiments of the inventive concepts the first conductive materials **211** to **291** may include or may be formed of a conductive material such as a polysilicon.
(99) The same structures as those on the first and second doping regions **311** and **312** may be provided in a region between the second and third doping regions **312** and **313**. In the region between the second and third doping regions **312** and **313**, a plurality of insulation materials **112** are provided, which extend along the first direction D**1**. A plurality of pillars **113** is provided that are disposed sequentially along the first direction D**1** and penetrate the plurality of insulation materials **112** along the third direction D**3**. An insulation layer **116** is provided on the exposed surfaces of the plurality of insulation materials **112** and the plurality of pillars **113**, and a plurality of first conductive materials **211** to **291** extend along the first direction D**1**. Similarly, the same structures as those on the first and second doping regions **311** and **312** may be provided in a region between the third and fourth doping regions **313** and **314**.
(100) A plurality of drain regions **321** are provided on the plurality of pillars **113**, respectively. The drain regions **321** may include or may be formed of silicon materials doped with a second type of charge carrier impurity. For example, the drain regions **321** may include or may be formed of silicon materials doped with an n-type dopant. In some example embodiments of the inventive concepts, the drain regions **321** include or are formed of n-type silicon materials. However, the drain regions **321** are not limited to n-type silicon materials.
(101) On the drain regions, a plurality of second conductive materials **331**, **332** and **333** are

provided, which extend along the second direction D**2**. The second conductive materials **331** to **333** are disposed along the first direction D**1**, being spaced apart from each other by a specific distance. The second conductive materials **331** to **333** are respectively connected to the drain regions **321** in a corresponding region. The drain regions **321** and the second conductive material **333** extending along the second direction D**2** may be connected through each contact plug. Each contact plug may be, for example, a conductive plug formed of a conductive material such as a metal. The second conductive materials **331** to **333** may include or may be formed of metal materials. The second conductive materials **331** to **333** may include or may be formed of conductive materials such as a polysilicon.

(102) In the example of FIG. **15**, the first conductive materials **211** to **291** may be used to form the wordlines WL, the string selection lines SSL and the ground selection lines GSL. For example, the first conductive materials **221** to **281** may be used to form the wordlines WL, where conductive materials belonging to the same layer may be interconnected. The second conductive materials **331** to **333** may be used to form the bitlines BL. The number of layers of the first conductive materials **211** to **291** may be changed variously according to process and control techniques.

(103) FIG. **16** is a circuit diagram illustrating an equivalent circuit of a memory block described with reference to FIG. **15** according to some example embodiments.

(104) A memory block BLKi shown in FIG. **16** may refer to a 3D memory block having a 3D structure formed on a substrate. For example, a plurality of memory NAND strings included in the memory block BLKi may be formed in a vertical direction to the substrate.

(105) Referring to FIG. **16**, the memory block BLKi may include a plurality of memory NAND strings (e.g., NS**11** to NS**33**), which are connected between bitlines BL**1**, BL**2**, and BL**3** and a common source line CSL. Each of the memory NAND strings NS**11** to NS**33** may include a string selection transistor SST, a plurality of memory cells (e.g., MC**1**, MC**2**, . . . , and MC**8**), and aground selection transistor GST. Each of the memory NAND strings NS**11** to NS**33** is illustrated as including eight memory cells MC**1**, MC**2**, . . . , and MC**8** in FIG. **15**, without being limited thereto.

(106) The string selection transistor SST may be connected to string selection lines SSL**1**, SSL**2**, and SSL**3** corresponding thereto. Each of the memory cells MC**1**, MC**2**, . . . , and MC**8** may be connected to a corresponding one of gate lines GTL**1**, GTL**2**, . . . , and/or GTL**8**. The gate lines GTL**1**, GTL**2**, . . . , and GTL**8** may respectively correspond to wordlines, and some of the gate lines GTL**1**, GTL**2**, . . . , and GTL**8** may correspond to dummy wordlines. The ground selection transistor GST may be connected to ground selection lines GSL**1**, GSL**2**, and GSL**3** corresponding thereto. The string selection transistor SST may be connected to the bitlines BL**1**, BL**2**, and BL**3** corresponding thereto, and the ground selection transistor GST may be connected to the common source line CSL.

(107) Wordlines (e.g., WL**1**) at the same level may be connected in common, and the ground selection lines GSL**1**, GSL**2**, and GSL**3** and the string selection lines SSL**1**, SSL**2**, and SSL**3** may be separated from each other. FIG. **16** illustrates example embodiments in which a memory block BLKi is connected to eight gate lines GTL**1**, GTL**2**, . . . , and GTL**8** and three bitlines BL**1**, BL**2**, and BL**3**, without being limited thereto.

(108) The number of the wordlines WL**1** to WL**8**, the number of the bitlines BL**1** to BL**3**, and the number of memory cells MC**1** to MC**8** are limited to an example of FIG. **16**.

(109) FIG. **17** is a cross-sectional view of a nonvolatile memory device included in a storage device according to some example embodiments.

(110) Referring to FIG. **17**, a nonvolatile memory device **2000** may have a chip-to-chip (C2C) structure. The C2C structure may refer to a structure formed by manufacturing an upper chip including a memory cell region or a cell region CELL on a first wafer, manufacturing a lower chip including a peripheral circuit region PERI on a second wafer, separate from the first wafer, and then bonding the upper chip and the lower chip to each other. Here, the bonding process may

include a method of electrically connecting a bonding metal formed on an uppermost metal layer of the upper chip and a bonding metal formed on an uppermost metal layer of the lower chip. For example, when the bonding metals may include copper (Cu) using a Cu-to-Cu bonding. The example embodiments, however, may not be limited thereto. For example, the bonding metals may also be formed of aluminum (Al) or tungsten (W).
(111) Each of the peripheral circuit region PERI and the cell region CELL of the nonvolatile memory device **2000** may include an external pad bonding area PA, a wordline bonding area WLBA, and a bitline bonding area BLBA.
(112) The peripheral circuit region PERI may include a first substrate **2210**, an interlayer insulating layer **2215**, a plurality of circuit elements **2220***a*, **2220***b*, and **2220***c* formed on the first substrate **2210**, first metal layers **2230***a*, **2230***b*, and **2230***c* respectively connected to the plurality of circuit elements **2220***a*, **2220***b*, and **2220***c*, and second metal layers **2240***a*, **2240***b*, and **2240***c* formed on the first metal layers **2230***a*, **2230***b*, and **2230***c*. In some example embodiments, the first metal layers **2230***a*, **2230***b*, and **2230***c* may be formed of tungsten having relatively high electrical resistivity, and the second metal layers **2240***a*, **2240***b*, and **2240***c* may be formed of copper having relatively low electrical resistivity.
(113) In example embodiments illustrated in FIG. **17**, although only the first metal layers **2230***a*, **2230***b*, and **2230***c* and the second metal layers **2240***a*, **2240***b*, and **2240***c* are shown and described, the example embodiments are not limited thereto, and one or more additional metal layers may be further formed on the second metal layers **2240***a*, **2240***b*, and **2240***c*. At least a portion of the one or more additional metal layers formed on the second metal layers **2240***a*, **2240***b*, and **2240***c* may be formed of aluminum or the like having a lower electrical resistivity than those of copper forming the second metal layers **2240***a*, **2240***b*, and **2240***c*.
(114) The interlayer insulating layer **2215** may be disposed on the first substrate **2210** and cover the plurality of circuit elements **2220***a*, **2220***b*, and **2220***c*, the first metal layers **2230***a*, **2230***b*, and **2230***c*, and the second metal layers **2240***a*, **2240***b*, and **2240***c*. The interlayer insulating layer **2215** may include an insulating material such as silicon oxide, silicon nitride, or the like.
(115) Lower bonding metals **2271***b* and **2272***b* may be formed on the second metal layer **2240***b* in the wordline bonding area WLBA. In the wordline bonding area WLBA, the lower bonding metals **2271***b* and **2272***b* in the peripheral circuit region PERI may be electrically bonded to upper bonding metals **2371***b* and **2372***b* of the cell region CELL. The lower bonding metals **2271***b* and **2272***b* and the upper bonding metals **2371***b* and **2372***b* may be formed of aluminum, copper, tungsten, or the like. Further, the upper bonding metals **2371***b* and **2372***b* in the cell region CELL may be referred as first metal pads and the lower bonding metals **2271***b* and **2272***b* in the peripheral circuit region PERI may be referred as second metal pads.
(116) The cell region CELL may include at least one memory block. The cell region CELL may include a second substrate **2310** and a common source line **2320**. On the second substrate **2310**, a plurality of wordlines **2331** to **2338** (e.g., **2330**) may be stacked in a third direction D**3** (e.g., a Z-axis direction), perpendicular to an upper surface of the second substrate **2310**. At least one string selection line and at least one ground selection line may be arranged on and below the plurality of wordlines **2330**, respectively, and the plurality of wordlines **2330** may be disposed between the at least one string selection line and the at least one ground selection line.
(117) In the bitline bonding area BLBA, a channel structure CH may extend in the third direction D**3** (e.g., the Z-axis direction), perpendicular to the upper surface of the second substrate **2310**, and pass through the plurality of wordlines **2330**, the at least one string selection line, and the at least one ground selection line. The channel structure CH may include a data storage layer, a channel layer, a buried insulating layer, and the like, and the channel layer may be electrically connected to a first metal layer **2350***c* and a second metal layer **2360***c*. For example, the first metal layer **2350***c* may be a bitline contact, and the second metal layer **2360***c* may be a bitline. In some example embodiments, the bitline **2360***c* may extend in a second direction D**2** (e.g., a Y-axis direction),

parallel to the upper surface of the second substrate **2310**.

(118) In example embodiments illustrated in FIG. **17**, an area in which the channel structure CH, the bitline **2360***c*, and the like are disposed may be defined as the bitline bonding area BLBA. In the bitline bonding area BLBA, the bitline **2360***c* may be electrically connected to the circuit elements **2220***c* providing a page buffer **2393** in the peripheral circuit region PERI. The bitline **2360***c* may be connected to upper bonding metals **2371***c* and **2372***c* in the cell region CELL, and the upper bonding metals **2371***c* and **2372***c* may be connected to lower bonding metals **2271***c* and **2272***c* connected to the circuit elements **2220***c* of the page buffer **2393**.

(119) In the wordline bonding area WLBA, the plurality of wordlines **2330** may extend in a first direction D**1** (e.g., an X-axis direction), parallel to the upper surface of the second substrate **2310** and perpendicular to the second direction D**2**, and may be connected to a plurality of cell contact plugs **2341** to **2347** (e.g., **2340**). The plurality of wordlines **2330** and the plurality of cell contact plugs **2340** may be connected to each other in pads provided by at least a portion of the plurality of wordlines **2330** extending in different lengths in the first direction D**1**. A first metal layer **2350***b* and a second metal layer **2360***b* may be connected to an upper portion of the plurality of cell contact plugs **2340** connected to the plurality of wordlines **2330**, sequentially. The plurality of cell contact plugs **2340** may be connected to the peripheral circuit region PERI by the upper bonding metals **2371***b* and **2372***b* of the cell region CELL and the lower bonding metals **2271***b* and **2272***b* of the peripheral circuit region PERI in the wordline bonding area WLBA. In some example embodiments, bonding metals **2251** and **2252** of the peripheral circuit area PERI may be connected to the cell area CELL through the bonding metal **2392** of the cell area CELL

(120) The plurality of cell contact plugs **2340** may be electrically connected to the circuit elements **2220***b* forming a row decoder **2394** in the peripheral circuit region PERI. In some example embodiments, operating voltages of the circuit elements **2220***b* forming the row decoder **2394** may be different than operating voltages of the circuit elements **2220***c* forming the page buffer **2393**. For example, operating voltages of the circuit elements **2220***c* forming the page buffer **2393** may be greater than operating voltages of the circuit elements **2220***b* forming the row decoder **2394**.

(121) A common source line contact plug **2380** may be disposed in the external pad bonding area PA. The common source line contact plug **2380** may be formed of a conductive material such as a metal, a metal compound, polysilicon, or the like, and may be electrically connected to the common source line **2320**. A first metal layer **2350***a* and a second metal layer **2360***a* may be stacked on an upper portion of the common source line contact plug **2380**, sequentially. For example, an area in which the common source line contact plug **2380**, the first metal layer **2350***a*, and the second metal layer **2360***a* are disposed may be defined as the external pad bonding area PA. The second metal layer **2360***a* may be electrically connected to the upper metal via **2371***a*. The upper metal via **2371***a* may be electrically connected to the upper metal pattern **2372***a*.

(122) Input/output pads **2205** and **2305** may be disposed in the external pad bonding area PA. A lower insulating film **2201** covering a lower surface of the first substrate **2210** may be formed below the first substrate **2210**, and a first input/output pad **2205** may be formed on the lower insulating film **2201**. The first input/output pad **2205** may be connected to at least one of the plurality of circuit elements **2220***a*, **2220***b*, and/or **2220***c* disposed in the peripheral circuit region PERI through a first input/output contact plug **2203**, and may be separated from the first substrate **2210** by the lower insulating film **2201**. In addition, a side insulating film may be disposed between the first input/output contact plug **2203** and the first substrate **2210** to electrically separate the first input/output contact plug **2203** and the first substrate **2210**.

(123) An upper insulating film **2301** covering the upper surface of the second substrate **2310** may be formed on the second substrate **2310**, and a second input/output pad **2305** may be disposed on the upper insulating film **2301**. The second input/output pad **2305** may be connected to at least one of the plurality of circuit elements **2220***a*, **2220***b*, and/or **2220***c* disposed in the peripheral circuit region PERI through a second input/output contact plug **2303**. In some example embodiments, the

second input/output pad **2305** is electrically connected to a circuit element **2220***a* disposed in the peripheral circuit area PERI through the second input and output contact plug **2303**, the lower metal pattern **2272***a*, and the lower metal via **2271***a*.
(124) According to some example embodiments, the second substrate **2310** and the common source line **2320** may not be disposed in an area in which the second input/output contact plug **2303** is disposed. Also, the second input/output pad **2305** may not overlap the wordlines **2330** in the third direction D**3** (e.g., the Z-axis direction). The second input/output contact plug **2303** may be separated from the second substrate **2310** in the direction, parallel to the upper surface of the second substrate **2310**, and may pass through the interlayer insulating layer **2315** of the cell region CELL to be connected to the second input/output pad **2305**.
(125) According to some example embodiments, the first input/output pad **2205** and the second input/output pad **2305** may be selectively formed. For example, the nonvolatile memory device **2000** may include only the first input/output pad **2205** disposed on the first substrate **2210** or the second input/output pad **2305** disposed on the second substrate **2310**. In some example embodiments, the nonvolatile memory device **2000** may include both the first input/output pad **2205** and the second input/output pad **2305**.
(126) A metal pattern provided on an uppermost metal layer may be provided as a dummy pattern or the uppermost metal layer may be absent, in each of the external pad bonding area PA and the bitline bonding area BLBA, respectively included in the cell region CELL and the peripheral circuit region PERI.
(127) In the external pad bonding area PA, the nonvolatile memory device **2000** may include a lower metal pattern **2273***a*, corresponding to an upper metal pattern **2372***a* formed in an uppermost metal layer of the cell region CELL, and having the same cross-sectional shape as the upper metal pattern **2372***a* of the cell region CELL so as to be connected to each other, in an uppermost metal layer of the peripheral circuit region PERI. In the peripheral circuit region PERI, the lower metal pattern **2273***a* formed in the uppermost metal layer of the peripheral circuit region PERI may not be connected to a contact. Similarly, in the external pad bonding area PA, an upper metal pattern **2372***a*, corresponding to the lower metal pattern **2273***a* formed in an uppermost metal layer of the peripheral circuit region PERI, and having the same shape as a lower metal pattern **2273***a* of the peripheral circuit region PERI, may be formed in an uppermost metal layer of the cell region CELL.
(128) The lower bonding metals **2271***b* and **2272***b* may be formed on the second metal layer **2240***b* in the wordline bonding area WLBA. In the wordline bonding area WLBA, the lower bonding metals **2271***b* and **2272***b* of the peripheral circuit region PERI may be electrically connected to the upper bonding metals **2371***b* and **2372***b* of the cell region CELL by a Cu-to-Cu bonding.
(129) Further, in the bitline bonding area BLBA, an upper metal pattern **2392**, corresponding to a lower metal pattern **2252** formed in the uppermost metal layer of the peripheral circuit region PERI, and having the same cross-sectional shape as the lower metal pattern **2252** of the peripheral circuit region PERI, may be formed in an uppermost metal layer of the cell region CELL. A contact may not be formed on the upper metal pattern **2392** formed in the uppermost metal layer of the cell region CELL.
(130) In some example embodiments, corresponding to a metal pattern formed in an uppermost metal layer in one of the cell region CELL and/or the peripheral circuit region PERI, a reinforcement metal pattern having the same cross-sectional shape as the metal pattern may be formed in an uppermost metal layer in the other one of the cell region CELL and/or the peripheral circuit region PERI. A contact may not be formed on the reinforcement metal pattern.
(131) The inventive concepts may be applied to various storage systems and computing systems including the storage systems. For example, the inventive concepts may be applied to systems such as a PC, a server computer, a data center, a workstation, a mobile phone, a smart phone, a tablet computer, a laptop computer, a PDA, a PMP, a digital camera, a portable game console, a music

player, a camcorder, a video player, a navigation device, a wearable device, an IoT device, an IoE device, an e-book reader, a VR device, an AR device, a robotic device, a drone, etc.

(132) As described herein, any devices, systems, blocks, modules, units, controllers, circuits, apparatus, and/or portions thereof according to any of some example embodiments (including, without limitation, any of the example embodiments of the computing system **100**, host **120**, host processor **140**, host memory **160**, storage system **200**, storage device **220**, storage device **240**, storage device **260**, write buffer **222**, write buffer **242**, first storage device **220***a*, host **120***a*, hypervisor **130***a*, storage device **220***a*, nonvolatile memory device **224***a*, first write buffer **222***a*, storage device **240***a*, nonvolatile memory device **244***a*, write buffer **242***a*, storage device **260***a*, storage device **220***b*, nonvolatile memory device **224***b*, first write buffer **222***b*, read buffer **228***b*, storage device **240***b*, nonvolatile memory device **244***b*, write buffer **242***b*, read buffer **248***b*, storage device **220***c*, nonvolatile memory device **244***c*, write buffer **242***c*, read buffer **248***c*, storage controller **223**, host interface **230**, CPU **233**, FTL **234**, packet manager **235**, memory interface **232**, AES engine **238**, ECC engine **237**, buffer memory **236**, nonvolatile memory device **224**, nonvolatille memory device **300**, control logic circuitry **320**, page buffer circuit **340**, voltage generator **350**, row decoder **360**, memory cell array **330**, any portion thereof, or the like) may include, may be included in, and/or may be implemented by one or more instances of processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a graphics processing unit (GPU), an application processor (AP), a digital signal processor (DSP), a microcomputer, a field programmable gate array (FPGA), and programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), a neural network processing unit (NPU), an Electronic Control Unit (ECU), an Image Signal Processor (ISP), and the like. In some example embodiments, the processing circuitry may include a non-transitory computer readable storage device (e.g., a memory), for example a solid state drive (SSD), storing a program of instructions, and a processor (e.g., CPU) configured to execute the program of instructions to implement the functionality and/or methods performed by some or all of any devices, systems, blocks, modules, units, controllers, circuits, apparatuses, and/or portions thereof according to any of some example embodiments, and/or any portions thereof, including for example some or all operations of any of the methods and/or processes shown in any of the drawings.

(133) The foregoing is illustrative of example embodiments and is not to be construed as limiting thereof. Although some example embodiments have been described, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from the novel teachings and advantages of the example embodiments. Accordingly, all such modifications are intended to be included within the scope of the example embodiments as defined in the claims. Therefore, it is to be understood that the foregoing is illustrative of various example embodiments and is not to be construed as limited to the specific example embodiments disclosed, and that modifications to the disclosed example embodiments, as well as some example embodiments, are intended to be included within the scope of the appended claims.

Claims

1. A storage system comprising: a first storage device including a first write buffer and a first nonvolatile memory device; and a second storage device including a second write buffer and a second nonvolatile memory device, wherein the first storage device is configured to, in response to a determination that a use buffer size of the first write buffer is greater than a first reference buffer size when the first storage device receives write data from a host, transfer the write data to the

second storage device, and the second storage device is configured to store the write data, wherein the second storage device is configured to, in response to a determination that the second storage device receives the write data from the first storage device, store the write data in the second write buffer, and write the write data stored in the second write buffer to a memory block of the second nonvolatile memory device, and wherein the first storage device further includes a mapping table, the second storage device is configured to transfer a logical address corresponding to an address of the memory block of the second nonvolatile memory device to the first storage device, and the first storage device is configured to change a physical address for the write data in the mapping table to the logical address received from the second storage device.
2. The storage system of claim 1, wherein the first storage device is configured to directly transfer the write data to the second storage device through peer-to-peer (P2P) communication.
3. The storage system of claim 1, wherein the first storage device is configured to, in response to a determination that the use buffer size of the first write buffer is decreased to less than a second reference buffer size, receive the write data from the second storage device, store the write data in the first write buffer, and write the write data stored in the first write buffer to the first nonvolatile memory device.
4. The storage system of claim 3, wherein the second storage device is configured to directly transfer the write data to the first storage device through P2P communication.
5. The storage system of claim 1, wherein the first storage device further includes a first read buffer, and the first storage device is configured to, in response to a determination that the first storage device receives a read command for the write data from the host concurrently with the write data being stored in the second storage device, receive the write data from the second storage device, store the write data in the first read buffer, and output the write data stored in the first read buffer to the host.
6. The storage system of claim 1, wherein the first storage device is configured to check a remaining buffer size of the second write buffer of the second storage device prior to transferring the write data to the second storage device.
7. The storage system of claim 6, wherein the first storage device is configured to transfer a buffer status request signal to the second storage device, and the second storage device is configured to transfer a buffer status response including the remaining buffer size of the second write buffer to the first storage device in response to the buffer status request signal.
8. The storage system of claim 7, wherein the buffer status response further includes information indicating a total buffer size of the second write buffer and information indicating whether the second write buffer is permitted to be used.
9. The storage system of claim 1, further comprising: at least one third storage device including a third write buffer, wherein the first storage device is configured to broadcast a buffer status request signal to the second storage device and the at least one third storage device through an interface bus, and wherein the second storage device and the at least one third storage device are configured to transfer buffer status responses for the second write buffer and the third write buffer through the interface bus in response to the buffer status request signal, respectively.
10. The storage system of claim 1, wherein the second storage device further includes a first read buffer, the first storage device is configured to request the write data to the second storage device with the logical address, and the second storage device is configured to store the write data in the first read buffer based on reading the write data from the memory block of the second nonvolatile memory device based on the address of the memory block of the second nonvolatile memory device corresponding to the logical address, and transfer the write data stored in the first read buffer to the first storage device.
11. A storage system, comprising: a first storage device including a first write buffer and a first nonvolatile memory device; and a second storage device including a second write buffer and a second nonvolatile memory device, wherein the first storage device is configured to, in response to

a determination that a use buffer size of the first write buffer is greater than a first reference buffer size when the first storage device receives write data from a host, transfer the write data to the second storage device, and the second storage device is configured to store the write data, wherein the second nonvolatile memory device includes a normal region that is accessible by the host, and an over-provisioning region that is not accessible by the host, and wherein the second storage device is configured to, in response to a determination that the second storage device receives the write data from the first storage device, store the write data in the second write buffer, and write the write data stored in the second write buffer to a memory block within the over-provisioning region.
12. The storage system of claim 11, wherein the first storage device includes a mapping table, the second storage device is configured to generate a logical address corresponding to an address of the memory block within the over-provisioning region, and transfer the logical address to the first storage device, and the first storage device is configured to change a physical address for the write data in the mapping table to the logical address received from the second storage device.
13. The storage system of claim 12, wherein the second storage device further includes a first read buffer, the first storage device is configured to request the write data to the second storage device with the logical address, and the second storage device is configured to store the write data in the first read buffer based on reading the write data from the memory block within the over-provisioning region based on the address of the memory block within the over-provisioning region corresponding to the logical address, and transfer the write data stored in the first read buffer to the first storage device.
14. A computing system, comprising: a storage system; and a host configured to store data in the storage system, wherein the storage system includes: a first storage device including a first write buffer and a first nonvolatile memory device; and a second storage device including a second write buffer and a second nonvolatile memory device, wherein the first storage device is configured to, in response to a determination that a use buffer size of the first write buffer is greater than a first reference buffer size when the first storage device receives write data from the host, transfer the write data to the second storage device, and the second storage device is configured to store the write data, wherein the second storage device is configured to, in response to a determination that the second storage device receives the write data from the first storage device, store the write data in the second write buffer, and write the write data stored in the second write buffer to a memory block of the second nonvolatile memory device, and wherein the first storage device further includes a mapping table, the second storage device is configured to transfer a logical address corresponding to an address of the memory block of the second nonvolatile memory device to the first storage device, and the first storage device is configured to change a physical address for the write data in the mapping table to the logical address received from the second storage device.